

Toward Safer Social Media Platforms

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ABSTRACT

Social media and the spreading of mobile devices have provided people with access to unprecedented opportunities anywhere and anytime. Yet, this has raised new challenges for users' privacy and safety, which become even more complex considering the different cultural values, norms and perceptiveness that people have regarding the correctness and safeness of online behaviours. In this context, this doctoral research project aims to investigate how social media platforms can empower their users, especially the vulnerable ones, and guarantee them a safer experience. In order to answer this question, I will exploit the interdisciplinary approach of the doctoral school and strong international connections, as well as the feedback I will collect from peers and experts by participating in conferences and in this Doctoral Consortium.

CCS CONCEPTS

• **Human-centered computing** → **Social networking sites; User interface programming**; • **Security and privacy** → **Usability in security and privacy**.

KEYWORDS

social media, messaging systems, online safety

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1 BACKGROUND AND MOTIVATION

People worldwide are being empowered by the spreading of social media and mobile devices. Indeed, they have significantly changed our lives, providing unprecedented opportunities and access to information and relationships anywhere and anytime [5]. On the other hand, this has raised several new concerns about users' safety and privacy, which are even more complex if we consider that the perception of privacy is not the same all over the world. One representative case study is represented by Arabic users, whose privacy perspective is influenced by Islamic teachings and culture, maintenance of reputation and the preservation of respect and modesty, whereas, in Western cultures, privacy is more focused

on concerns about safety and security [1]. Conception of privacy impacts how people use social media and interact with others online and through mobile devices, by requiring designers and developers to consider cultural values while building social media platforms. In this context, Fisher [7] discussed the design of social media and mobile tools to support and empower Syrian women refugees. The first step towards safer and more trustable use of social media based on the Arab culture has been made in [8]. The interest of the scientific community in considering (Arab) cultural values in the use of technology, and hence even in the design of social media platforms, to ensure their adaptability and success, is also evidenced by initiatives such as ArabHCI [14].

Furthermore, there is also a difference between the safety of messaging applications and social media platforms and the one perceived by users, which sometimes is not coherent with the features provided by the application [9]. For instance, this happens in the case of sexting, the practice of sending or receiving self-generated sexual content (e.g., text, images, videos, etc.) through social media platforms (e.g., social networks, dating applications, messaging applications, etc.) [2]. Although its popularity among (but not only) teenagers [18], it carries dangers such as the possible spread of personal sexual-related images without the consent of the owner, which can bring significant damage to the victims, even effects similar to the ones of physical violence [3, 20]. In this context, commonly used messaging applications are not safe enough for sexting since they permit sending any content to anyone without any limitation, thus exposing their users to the issues related to sexting abuse [9]. As a first attempt toward safer messaging environments, in [9], I proposed and discussed some guidelines for developers to build safer-by-design social media platforms. Some suggestions on how to design peer support platforms and new messaging platforms taking into account teens worries related to sexting are also reported in [13]. Instead, a literature review of computational approaches for online sexual risk detection from a human-centered perspective is presented in [19].

Besides these two examples, the safe and trustable use of social media is a complex and broad concept which includes several aspects. Indeed, several works discussed these topics from different points of view. For instance, some papers addressed the topic of the safe use of mobile devices and social networks by children and teenagers, discussing the design of parental control applications and acknowledging that a permissive approach is more effective than a restrictive one [10, 17, 23]. Halevy *et al.* [12] discussed the problem of the preservation of integrity in online social networks, i.e. keeping online platforms and their users safe from malicious activities, presenting recent progress and ongoing challenges. Instead, Liu *et al.* [16] presented a review regarding image privacy in online social networks, describing the issues of every stage of image

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sharing, from the creation to the viewing and the social part, even highlighting the open research challenges in the field. Li *et al.* [15] proposed a taxonomy of content sensitivity by employing a new method to collect sensitive images. By exploiting this new method, Vishwamitra *et al.* [22] collected images to train the proposed deep learning model. Indeed, they presented a novel system to enable automatic and user-specific content-based photo privacy control in online social networks. They also described a possible end-to-end flow where their proposal can be employed.

Yet, the road to the diffusion and broad adoption of new and visionary messaging platforms and social media is full of obstacles. Amongst the challenges that users face when trying to switch from one messaging application to another (e.g., learning how to use a new application, different or missing functionalities, lack of trust, breaking habits, etc.), network effects are the most common [11]. In other words, moving from one messaging platform to another is not an individual choice but a social one. A possible yet unexplored solution to the negative consequences of network effects is interoperability. The necessity for interoperable messaging environments is also pushed by recent European regulations, in particular the “European Electronic Communications Code” (ECC) and the “Digital Markets Act” (DMA). In particular, the latter will impose big tech companies (also called gatekeepers) to make their messaging (but not only) systems interoperable with those of smaller businesses [4]. Besides the technical challenges in designing such messaging protocols, this poses new and mostly unexplored questions for HCI researchers. For instance, the minimum set of features that could be considered sufficient by users for an interoperable messaging environment is yet unknown.

Last but not least, education, in the long term, is the main weapon in preventing online-related risks and toward responsible and safe use of social media. Since digital literacy and digital citizenship are becoming increasingly important, several initiatives and tools have been created to address such topics with people of various ages. Some representative examples are the Common Sense Education K-12 Digital Citizenship Curriculum, which offers educational material on many digital citizenship topics, and Facebook’s Messenger Kids, an alternative to the well-known Facebook Messenger, which allows parents to monitor their children. Yet, most of the available tools do not have a social media component. Instead, hands-on experiences would be more effective in developing lifelong skills. In this direction, DiFranzo *et al.* [6] presented “Social Media TestDrive”, an interactive social media simulation which combines experimental learning through a safe social media environment and classroom lessons. However, this tool is designed only with teenagers in mind, whereas a more comprehensive approach is needed. Furthermore, it is just a simulation, thus making difficult the full integration in a real social media platform.

2 RESEARCH GOALS AND APPROACHES

In this context, the overall goal of this doctoral project is to answer the following research question:

How can social media platforms empower their users, especially vulnerable ones, and guarantee them a safer experience?

More practically, the aim of this project is to investigate solutions that could be used to build safer social media platforms (e.g., forwarding control of personal content, trigger warnings considering different cultures, etc.), also showing the potentialities and the limits of current technologies in protecting users of messaging platforms from online related risks and abuses. Tradeoffs between the effectiveness of privacy-preserving features and the usability of the system will also be considered and discussed. Furthermore, as a possible solution to network effects, the design of interoperable messaging environments will be analyzed from a human-centered perspective, even developing solutions and testing them with real users. Part of this project will also focus on improving users’ awareness and literacy about online-related risks and knowledge about correct and safe behaviours on social media and through mobile devices. To summarize, some problems that will be addressed by this work are:

- (1) design and develop a messaging system safer by design for sexting, analyzing possible solutions, as well as their limits, in creating a social environment where users retain some form of control over their sensitive self-generated content;
- (2) propose privacy-preserving features for a messaging system or a social network considering the peculiarities of the Arab culture, or, more in general, a highly-personalized social environment able to adapt itself considering several factors (e.g., culture of the users, etc.);
- (3) understand how to design an interoperable messaging environment from a human-centered perspective and evaluate it considering both its performance and its usability;
- (4) design solutions (e.g., features, serious games, etc.) to improve users’ awareness and digital literacy so that their exposure to risks online and on social media can be reduced.

To the best of my knowledge, this is the first attempt in these directions.

The impact of this research project is expected to be both in the short and long term. Indeed, the proposed features for safer social media platforms can have an immediate positive impact as soon as they are released. On the other hand, improving people’s digital literacy and knowledge is a long-term process. Furthermore, by exposing their negligence, I also expect to have an impact on Internet companies and service providers, and through the proposed solutions and proofs-of-concept, I hope to stimulate them to design their systems considering users’ safety, privacy, and control over personal content.

Several tools and approaches will be employed to reach the goals mentioned above, depending on the specific topic and need. As stated in Section 3, the outcomes of a questionnaire are discussed in [9]. Thus, surveys are a tool used in this research project. Yet, to reduce or remove the response bias, even analyses of data collected from social networks can be performed. For instance, posts and comments gathered in a teen mental health support forum are employed in [18] and [13], showing that it is an approach used in the field. The work presented in [8] is based on the results of some workshops held in Jordan. As discussed in Section 3, this doctoral project incorporates the work described in that paper. Semi-structured interviews could be used as well.

3 RESULTS AND CONTRIBUTIONS TO DATE

As mentioned above, the overall goal of this doctoral project is to investigate technologies able to create a safer messaging platform (or, more in general, social media) with an interdisciplinary approach. In this context, I have already contributed to two works discussing the first steps toward the aims of this research project.

In [9], I discussed the relationships between sexting and disability. Indeed, I argued how, although the well-known risks related to this practice, it represents an opportunity for people with impairments to explore their sexuality in an exposure-controlled way. However, through a questionnaire administered to some master's students of Computer Science and their contacts, I discovered that the perception of the sexuality (and, as a consequence, of the practice of sexting) of people with disabilities is fraught with prejudice. I also observed that people have an optimistic view of themselves, evaluating their sexting-related risk lower than what their claim for their age group. Besides that, I proposed some guidelines for developers to build safer messaging platforms that could allow some form of media forwarding control. The proposed system is based on a perceptual hashing function and a forwarding control algorithm able to detect when a suspicious forwarding is about to happen. I have submitted an extended paper that presents this system in detail to the IEEE Transactions on Mobile Computing journal (currently under review).

I am also exploring some technologies to design a messaging system (or, more in general, a social media) based on the Arab culture thanks to a collaboration with an international team which includes Prof. Karen E. Fisher from the University of Washington (USA), who is currently spending a research period in Jordan, and Prof. Yacine Ghamri-Doudane from the University of La Rochelle (France). The first outcome of this collaboration is presented in [8]. In that work, starting from ongoing fieldwork in Jordan, I identified some privacy-preserving features that should be embedded in messaging systems considering the Arab culture, even though they are useful for any user. I also discussed some possible technological approaches to implement them and proposed a proof-of-concept of a messaging platform endowed with a feature to detect malicious URLs. I have now moved my focus on hiding sensitive content, as disturbing content is a potential cause of trauma [21], considering the various cultures, aiming to design a social media able to adapt itself to the single user.

4 ACADEMIC STATUS

I am currently enrolled in the first year of the doctoral program in Brain, Mind and Computer Science (BMCS) at the University of Padua (Italy), under the supervision of Prof. Claudio E. Palazzi. Although this PhD program is part of the Department of General Psychology, it emerges from the close collaboration between faculties of cognitive neuroscience, computer science and psychology around the unifying topic of Human-Computer Interaction (HCI). Indeed, it proposes two curricula: (1) Neuroscience, Technology, and Society, and (2) Computer Science and Innovation for Societal Challenges. Each student is affiliated with one of them, and must have a main supervisor from its curricula and a co-supervisor from the other one, fostering an interdisciplinary approach. Exploiting the peculiarities of this PhD program, my doctoral project is highly

interdisciplinary, focusing on mobile computing, social networks and HCI. In particular, the last two require the expertise of psychologists and sociologists. Some parts of the project (e.g., improving digital literacy) have indeed been specifically designed to foster collaborations with them.

The duration of the course is three years (full-time). Therefore, I am expected to graduate in 2026. Since I am a first-year student, I have never participated in a Doctoral Consortium. Yet, I started this project in May 2021 for my Master Thesis (discussed in December 2021) and continued till now, first with a fellowship granted by the University of Padua and then as Research Engineer at the University of La Rochelle. I am hence in the perfect time frame for a doctoral consortium, as my research project has already been going on for more than one year, and it will last for another 3 years.

5 NEXT STEPS

Although I have already made some steps towards the goal of this research project, there are still many directions in which it can be expanded. I have already planned some of the next steps, whereas I am discussing others. For instance, I am about to start a collaboration with Prof. Barbara Guidi (University of Pisa, Italy), aiming to investigate how decentralized architectures (e.g., blockchain) can be employed to build a safer messaging environment, considering sexting as a case study. Furthermore, I am keeping the ongoing collaboration with Prof. Yacine Ghamri-Doudane (University of La Rochelle, France) and Prof. Karen E. Fisher (University of Washington, USA) to design social media suitable even for Arab citizens and improve the digital literacy of Syrian women refugees, drawing from ongoing fieldwork in Jordan. These collaborations not only increase the potential impact of my research but also make it more interdisciplinary and international. In addition, exploiting the interdisciplinary nature of this doctoral program, I plan to create collaborations with psychologists of this doctoral school (but not only), both faculties and PhD students. For instance, the design of serious games, functionalities and solutions to improve users' awareness on online-related risks, even considering vulnerable people (e.g., people with low digital literacy, children), can be a good topic also for such collaborations.

Finally, research becomes good research also thanks to the feedback received from the scientific community. I plan to disseminate the outcomes of this doctoral project, publishing in top-tier international conferences and journals, such as Communications of the ACM, IEEE Transactions on Mobile Computing, International Conference on Human Factors in Computing Systems (CHI), etc. This will allow me to improve the quality of my research thanks to the feedback received by the referees. Then, publishing and participating in conferences will also permit me to share ideas, collect feedback from the other participants, and create new international collaborations. This project can indeed be adjusted considering the feedback and ideas that I will collect, for instance, during this Doctoral Consortium.

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